

727723EUA1109 : SANJAY M

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import mean_squared_error
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
from tensorflow.keras.callbacks import EarlyStopping

np.random.seed(42)
tf.random.set_seed(42)

t = np.linspace(0, 6*np.pi, 300)
y = np.sin(t) + np.random.normal(0, 0.1, 300)

INPUT_STEPS, OUTPUT_STEPS = 10, 3

X, Y = [], []
for i in range(len(y) - INPUT_STEPS - OUTPUT_STEPS + 1):
    X.append(y[i:i+INPUT_STEPS])
    Y.append(y[i+INPUT_STEPS:i+INPUT_STEPS+OUTPUT_STEPS])

X, Y = np.array(X)[..., np.newaxis], np.array(Y)
print(f"X.shape: {X.shape}, Y.shape: {Y.shape}")

split = int(0.8 * len(X))
X_train, X_test = X[:split], X[split:]
y_train, y_test = Y[:split], Y[split:]
print(f"Train: {X_train.shape}, Test: {X_test.shape}")
```

```
X.shape: (288, 10, 1), Y.shape: (288, 3)
Train: (230, 10, 1), Test: (58, 10, 1)
```

```
In [2]: def build_model():
    model = Sequential([
        LSTM(64, input_shape=(INPUT_STEPS, 1)),
        Dense(32, activation='relu'),
        Dense(OUTPUT_STEPS)
    ])
    model.compile(optimizer='adam', loss='mse')
    return model

model = build_model()
model.summary()

es = EarlyStopping(patience=10, restore_best_weights=True)
history = model.fit(X_train, y_train, epochs=150, batch_size=32,
                    validation_split=0.1, callbacks=[es], verbose=1)
```

```
c:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\rnn\rnn.py:199: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
```

```
super().__init__(**kwargs)
```

```
Model: "sequential"
```

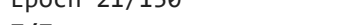
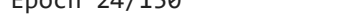
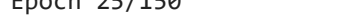
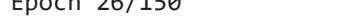
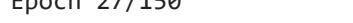
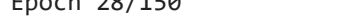
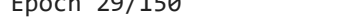
Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 64)	16,896
dense (Dense)	(None, 32)	2,080
dense_1 (Dense)	(None, 3)	99



Total params: 19,075 (74.51 KB)

Trainable params: 19,075 (74.51 KB)

Non-trainable params: 0 (0.00 B)

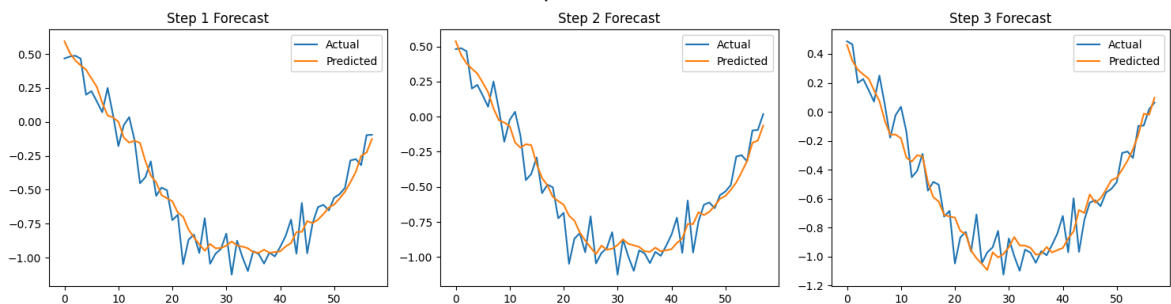
Epoch 1/150
7/7  3s 72ms/step - loss: 0.4541 - val_loss: 0.4687
Epoch 2/150
7/7  0s 18ms/step - loss: 0.3158 - val_loss: 0.1944
Epoch 3/150
7/7  0s 17ms/step - loss: 0.2005 - val_loss: 0.0475
Epoch 4/150
7/7  0s 19ms/step - loss: 0.1359 - val_loss: 0.0412
Epoch 5/150
7/7  0s 18ms/step - loss: 0.0971 - val_loss: 0.0332
Epoch 6/150
7/7  0s 19ms/step - loss: 0.0765 - val_loss: 0.0409
Epoch 7/150
7/7  0s 17ms/step - loss: 0.0702 - val_loss: 0.0296
Epoch 8/150
7/7  0s 18ms/step - loss: 0.0639 - val_loss: 0.0266
Epoch 9/150
7/7  0s 18ms/step - loss: 0.0578 - val_loss: 0.0268
Epoch 10/150
7/7  0s 21ms/step - loss: 0.0516 - val_loss: 0.0280
Epoch 11/150
7/7  0s 21ms/step - loss: 0.0453 - val_loss: 0.0250
Epoch 12/150
7/7  0s 21ms/step - loss: 0.0384 - val_loss: 0.0239
Epoch 13/150
7/7  0s 20ms/step - loss: 0.0315 - val_loss: 0.0236
Epoch 14/150
7/7  0s 43ms/step - loss: 0.0248 - val_loss: 0.0216
Epoch 15/150
7/7  0s 30ms/step - loss: 0.0198 - val_loss: 0.0192
Epoch 16/150
7/7  0s 28ms/step - loss: 0.0168 - val_loss: 0.0185
Epoch 17/150
7/7  0s 19ms/step - loss: 0.0159 - val_loss: 0.0182
Epoch 18/150
7/7  0s 21ms/step - loss: 0.0154 - val_loss: 0.0181
Epoch 19/150
7/7  0s 21ms/step - loss: 0.0150 - val_loss: 0.0181
Epoch 20/150
7/7  0s 20ms/step - loss: 0.0149 - val_loss: 0.0181
Epoch 21/150
7/7  0s 18ms/step - loss: 0.0147 - val_loss: 0.0182
Epoch 22/150
7/7  0s 22ms/step - loss: 0.0145 - val_loss: 0.0184
Epoch 23/150
7/7  0s 19ms/step - loss: 0.0144 - val_loss: 0.0184
Epoch 24/150
7/7  0s 21ms/step - loss: 0.0144 - val_loss: 0.0183
Epoch 25/150
7/7  0s 20ms/step - loss: 0.0143 - val_loss: 0.0183
Epoch 26/150
7/7  0s 21ms/step - loss: 0.0142 - val_loss: 0.0183
Epoch 27/150
7/7  0s 22ms/step - loss: 0.0141 - val_loss: 0.0182
Epoch 28/150
7/7  0s 20ms/step - loss: 0.0141 - val_loss: 0.0182
Epoch 29/150
7/7  0s 23ms/step - loss: 0.0141 - val_loss: 0.0182

```
In [3]: y_pred = model.predict(X_test)

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
for i, ax in enumerate(axes):
    ax.plot(y_test[:, i], label='Actual')
    ax.plot(y_pred[:, i], label='Predicted')
    ax.set_title(f'Step {i+1} Forecast')
    ax.legend()
plt.tight_layout()
plt.show()

for i in range(OUTPUT_STEPS):
    mse = mean_squared_error(y_test[:, i], y_pred[:, i])
    print(f'Step {i+1} MSE: {mse:.6f}')
```

2/2 ————— 0s 261ms/step



Step 1 MSE: 0.015703

Step 2 MSE: 0.014411

Step 3 MSE: 0.013973

```
In [4]: def build_deep_model():
    model = Sequential([
        LSTM(64, return_sequences=True, input_shape=(INPUT_STEPS, 1)),
        LSTM(64),
        Dense(32, activation='relu'),
        Dense(OUTPUT_STEPS)
    ])
    model.compile(optimizer='adam', loss='mse')
    return model

deep_model = build_deep_model()
deep_model.summary()

es2 = EarlyStopping(patience=10, restore_best_weights=True)
deep_model.fit(X_train, y_train, epochs=150, batch_size=32,
               validation_split=0.1, callbacks=[es2], verbose=1)

y_pred_deep = deep_model.predict(X_test)

print("\n--- Shallow vs Deep MSE Comparison ---")
for i in range(OUTPUT_STEPS):
    mse_shallow = mean_squared_error(y_test[:, i], y_pred[:, i])
    mse_deep = mean_squared_error(y_test[:, i], y_pred_deep[:, i])
    print(f'Step {i+1} | Shallow MSE: {mse_shallow:.6f} | Deep MSE: {mse_deep:.6f}')
```

c:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\rnn\rnn.py:199: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

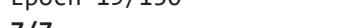
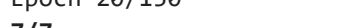
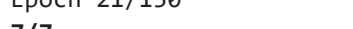
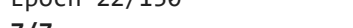
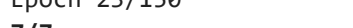
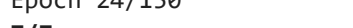
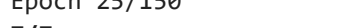
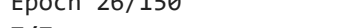
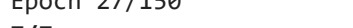
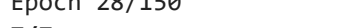
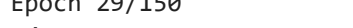
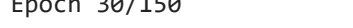
```
super().__init__(**kwargs)
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
lstm_1 (LSTM)	(None, 10, 64)	16,896
lstm_2 (LSTM)	(None, 64)	33,024
dense_2 (Dense)	(None, 32)	2,080
dense_3 (Dense)	(None, 3)	99



Total params: 52,099 (203.51 KB)
Trainable params: 52,099 (203.51 KB)
Non-trainable params: 0 (0.00 B)

Epoch 1/150
7/7  8s 131ms/step - loss: 0.4439 - val_loss: 0.6863
Epoch 2/150
7/7  0s 37ms/step - loss: 0.3064 - val_loss: 0.5333
Epoch 3/150
7/7  0s 49ms/step - loss: 0.2161 - val_loss: 0.2094
Epoch 4/150
7/7  0s 29ms/step - loss: 0.1215 - val_loss: 0.0477
Epoch 5/150
7/7  0s 41ms/step - loss: 0.1325 - val_loss: 0.0346
Epoch 6/150
7/7  0s 39ms/step - loss: 0.1114 - val_loss: 0.0671
Epoch 7/150
7/7  0s 37ms/step - loss: 0.1013 - val_loss: 0.0440
Epoch 8/150
7/7  0s 42ms/step - loss: 0.0914 - val_loss: 0.0333
Epoch 9/150
7/7  0s 27ms/step - loss: 0.0863 - val_loss: 0.0325
Epoch 10/150
7/7  0s 40ms/step - loss: 0.0765 - val_loss: 0.0312
Epoch 11/150
7/7  0s 37ms/step - loss: 0.0646 - val_loss: 0.0276
Epoch 12/150
7/7  0s 36ms/step - loss: 0.0529 - val_loss: 0.0245
Epoch 13/150
7/7  0s 38ms/step - loss: 0.0388 - val_loss: 0.0221
Epoch 14/150
7/7  0s 43ms/step - loss: 0.0266 - val_loss: 0.0297
Epoch 15/150
7/7  0s 25ms/step - loss: 0.0225 - val_loss: 0.0315
Epoch 16/150
7/7  0s 39ms/step - loss: 0.0207 - val_loss: 0.0290
Epoch 17/150
7/7  0s 41ms/step - loss: 0.0190 - val_loss: 0.0284
Epoch 18/150
7/7  0s 27ms/step - loss: 0.0192 - val_loss: 0.0199
Epoch 19/150
7/7  0s 39ms/step - loss: 0.0162 - val_loss: 0.0188
Epoch 20/150
7/7  0s 33ms/step - loss: 0.0147 - val_loss: 0.0184
Epoch 21/150
7/7  0s 30ms/step - loss: 0.0149 - val_loss: 0.0192
Epoch 22/150
7/7  0s 41ms/step - loss: 0.0152 - val_loss: 0.0190
Epoch 23/150
7/7  0s 38ms/step - loss: 0.0150 - val_loss: 0.0181
Epoch 24/150
7/7  0s 38ms/step - loss: 0.0145 - val_loss: 0.0179
Epoch 25/150
7/7  0s 36ms/step - loss: 0.0143 - val_loss: 0.0179
Epoch 26/150
7/7  0s 37ms/step - loss: 0.0144 - val_loss: 0.0181
Epoch 27/150
7/7  0s 40ms/step - loss: 0.0145 - val_loss: 0.0179
Epoch 28/150
7/7  0s 37ms/step - loss: 0.0143 - val_loss: 0.0178
Epoch 29/150
7/7  0s 35ms/step - loss: 0.0142 - val_loss: 0.0177
Epoch 30/150
7/7  0s 36ms/step - loss: 0.0141 - val_loss: 0.0177

Epoch 31/150
7/7  0s 45ms/step - loss: 0.0141 - val_loss: 0.0177

Epoch 32/150
7/7  0s 43ms/step - loss: 0.0141 - val_loss: 0.0175

Epoch 33/150
7/7  0s 36ms/step - loss: 0.0139 - val_loss: 0.0175

Epoch 34/150
7/7  0s 45ms/step - loss: 0.0139 - val_loss: 0.0175

Epoch 35/150
7/7  0s 34ms/step - loss: 0.0138 - val_loss: 0.0174

Epoch 36/150
7/7  0s 30ms/step - loss: 0.0138 - val_loss: 0.0175

Epoch 37/150
7/7  0s 24ms/step - loss: 0.0139 - val_loss: 0.0174

Epoch 38/150
7/7  0s 19ms/step - loss: 0.0138 - val_loss: 0.0173

Epoch 39/150
7/7  0s 21ms/step - loss: 0.0137 - val_loss: 0.0173

Epoch 40/150
7/7  0s 17ms/step - loss: 0.0137 - val_loss: 0.0173

Epoch 41/150
7/7  0s 16ms/step - loss: 0.0137 - val_loss: 0.0173

Epoch 42/150
7/7  0s 16ms/step - loss: 0.0137 - val_loss: 0.0173

Epoch 43/150
7/7  0s 17ms/step - loss: 0.0137 - val_loss: 0.0172

Epoch 44/150
7/7  0s 22ms/step - loss: 0.0137 - val_loss: 0.0172

Epoch 45/150
7/7  0s 17ms/step - loss: 0.0136 - val_loss: 0.0171

Epoch 46/150
7/7  0s 20ms/step - loss: 0.0136 - val_loss: 0.0172

Epoch 47/150
7/7  0s 17ms/step - loss: 0.0137 - val_loss: 0.0172

Epoch 48/150
7/7  0s 17ms/step - loss: 0.0136 - val_loss: 0.0172

Epoch 49/150
7/7  0s 16ms/step - loss: 0.0136 - val_loss: 0.0171

Epoch 50/150
7/7  0s 18ms/step - loss: 0.0136 - val_loss: 0.0170

Epoch 51/150
7/7  0s 17ms/step - loss: 0.0135 - val_loss: 0.0170

Epoch 52/150
7/7  0s 17ms/step - loss: 0.0135 - val_loss: 0.0171

Epoch 53/150
7/7  0s 17ms/step - loss: 0.0136 - val_loss: 0.0170

Epoch 54/150
7/7  0s 18ms/step - loss: 0.0135 - val_loss: 0.0170

Epoch 55/150
7/7  0s 18ms/step - loss: 0.0135 - val_loss: 0.0170

Epoch 56/150
7/7  0s 17ms/step - loss: 0.0135 - val_loss: 0.0170

Epoch 57/150
7/7  0s 20ms/step - loss: 0.0135 - val_loss: 0.0170

Epoch 58/150
7/7  0s 31ms/step - loss: 0.0135 - val_loss: 0.0171

Epoch 59/150
7/7  0s 29ms/step - loss: 0.0135 - val_loss: 0.0170

Epoch 60/150
7/7  0s 37ms/step - loss: 0.0134 - val_loss: 0.0170

Epoch 61/150
7/7  0s 30ms/step - loss: 0.0134 - val_loss: 0.0170
Epoch 62/150
7/7  0s 29ms/step - loss: 0.0134 - val_loss: 0.0171
Epoch 63/150
7/7  0s 26ms/step - loss: 0.0135 - val_loss: 0.0170
Epoch 64/150
7/7  0s 25ms/step - loss: 0.0134 - val_loss: 0.0170
2/2  1s 393ms/step

--- Shallow vs Deep MSE Comparison ---

Step 1	Shallow MSE: 0.015703	Deep MSE: 0.014837
Step 2	Shallow MSE: 0.014411	Deep MSE: 0.014324
Step 3	Shallow MSE: 0.013973	Deep MSE: 0.015532